

$$x = 0.11 \text{ m}$$

Q7 Ans: B

$$\begin{aligned}\text{Work done by gravity} &= mg \times \text{vertical displacement in direction of } mg \\ &= 990 \times 9.81 \times (-12 \times 8.3) \times \sin 15^\circ \\ &= -250 \text{ kJ}\end{aligned}$$

Q8 Ans: A